0580_w24_ms_42_Q3

Question	Answer	Marks	Partial Marks	Guidance
3(a)(i)	$-m + 3n$ final answer	**2**	**B1** for $-m$ or [+] $3n$ in final answer or for $-m + 3n$ seen and then spoiled	Accept eg $3n - 1m$
3(a)(ii)	$81a^8c^{12}$ final answer	**2**	**B1** for final answer in correct form with any two of 81, a^8, c^{12} correct or for $81a^8c^{12}$ seen and then spoiled	
3(a)(iii)	$\frac{19x}{30}$ final answer	**2**	**M1** for $\frac{6\times4x-3\times3x+2\times2x}{30}$ oe	M1 e.g. $\frac{12x-4.5x+2x}{15}$ or answer $\frac{9.5x}{15}$ For M1 condone e.g. $\frac{24x-13x}{30}$ oe if addition done first , may be implied by answer $\frac{11x}{30}$ oe
3(b)	4.5 oe	**3**	**M1** for $a + 2(3a + 2) = 35.5$ oe **M1** for correct $ka = b$ for *their* linear equation	
3(c)	$\frac{-(-4)\pm\sqrt{(-4)^2-4\times5\times(-3)}}{2\times5}$ oe	**M2**	**M1** for $\sqrt{(-4)^2-4\times5\times(-3)}$ or better or for $\frac{-(-4)+\sqrt{q}}{2\times5}$ or $\frac{-(-4)-\sqrt{q}}{2\times5}$ or better	For M2 allow $\frac{4\pm\sqrt{76}}{10}$ but not $\frac{2\pm\sqrt{19}}{5}$ **No recovery for short line or short root or -4^2 unless seen or implied in working** Fraction line must underscore the ± sign Length of root must cover – sign, ie reach the second 4
	–0.472 or –0.4718 to –0.4717 **and** 1.27 or 1.271 to 1.272	**B1**		Mark at most accurate for decimal answers Condone answers $\frac{2+\sqrt{19}}{5}$, $\frac{2-\sqrt{19}}{5}$ for B1

Question	Answer	Marks	Partial Marks	Guidance
3(d)	$x^2 - 6x + 8$ [= 0] or $y^2 - 6y + 5$ [= 0]	**M2**	**M1** for $x^2 - 4x + 5 = 2x - 3$ or $y = \left(\frac{y+3}{2}\right)^2 - 4\left(\frac{y+3}{2}\right) + 5$	M2 accept if signs all reversed e.g. $-x^2 + 6x - 8$ [= 0]
	$(x - 4)(x - 2)$ [= 0] or $(y - 1)(y - 5)$ [=0] OR [x =] $\frac{-(-6) \pm \sqrt{(-6)^2 - 4[\times 1] \times 8}}{2[\times 1]}$ or [y =] $\frac{-(-6) \pm \sqrt{(-6)^2 - 4[\times 1] \times 5}}{2[\times 1]}$ OR [x =] $3 \pm \sqrt{-8+9}$ or [y =] $3 \pm \sqrt{-5+9}$	**M1**	FT *their* 3-term quadratic but not if $x^2 - 4x + 5$[= 0]	For M1 condone partially factorised e.g. $x(x - 2) - 4(x - 2)$ if followed by correct values of x for *their* quadratic. If they do not show full working accept e.g. $\frac{6 \pm \sqrt{4}}{2}$ as minimum working for M1. If completing the square, they must make x the subject Working must be shown for this M1, solutions alone is M0
	[x =] 2 , [y =] 1 [x =] 4 , [y =] 5	**B2**	**B1** for one correct pair or two correct x-values or two correct y-values If B0 scored **and** at least 2 method marks scored, **SC1** for correct substitution of both of *their x* values or *their y* values into $y = x^2 - 4x + 5$ or $y = 2x - 3$	For SC1, the substitution must be shown